

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Perihalan Produk:

Product Description:

Cat No. :

Pasteurella multocida Toxin PMT Kit 1 plate

Pasteurella multocida Toxin PMT Kit 1 plate

K000911-9

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use

In vitro diagnostic.

Uses advised against

No Information available

Details of the supplier of the safety data sheet

Company

Thermo Scientific Microbiology Sdn Bhd
No.6, Jalan TTC 6, Taman Teknologi Cheng,
Cheng, 75250 Melaka, Malaysia
+606 334 0975 .

Supplier

Oxoid Ltd.
Wade Road
Basingstoke, Hants, UK
RG24 8PW
Telephone: +44 (0) 1256 841144

E-mail address

mbd-sds@thermofisher.com

Emergency Telephone Number

(603) 5122 8888
CHEMTREC Malaysia **1-800-815-308** (Malay)
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Skin Corrosion/Irritation	Category 2 (H315)
Serious Eye Damage/Eye Irritation	Category 2 (H319)
Reproductive Toxicity	Category 1B (H360D)

Label Elements



Signal Word

Danger

Hazard Statements

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H315 - Causes skin irritation
H319 - Causes serious eye irritation
H360D - May damage the unborn child

Precautionary Statements

P332 + P313 - If skin irritation occurs: Get medical advice/attention
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P280 - Wear protective gloves/protective clothing/eye protection/face protection

Other Hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS No	Weight %
WASH BUFFER - Tris-hydrochloride	1185-53-1	< 20.0
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	872-50-4	5 - 10
Propylene carbonate	108-32-7	10 - 25
STOP SOLUTION - Sulphuric acid	7664-93-9	4.5
Tris (hydroxymethyl) aminomethane	77-86-1	1 - 5
1-Aminopyrine	58-15-1	0.08

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

If symptoms persist, call a physician.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention.

Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

Most important symptoms and effects, both acute and delayed

Irritating to eyes. Irritating to skin.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

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SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

None under normal use conditions.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing.

Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed. Keep at temperatures between 2° and 8 °C.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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Control Parameters

Component	Malaysia	ACGIH TLV	OSHA PEL
STOP SOLUTION - Sulphuric acid		TWA: 0.2 mg/m ³	(Vacated) TWA: 1 mg/m ³ TWA: 1 mg/m ³

Component	European Union	The United Kingdom	Germany
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	TWA: 40 mg/m ³ (8h) TWA: 10 ppm (8h) Skin STEL: 20 ppm (15min) STEL: 80 mg/m ³ (15min) STEL: 80 mg/m ³ (8h) STEL: 20 ppm (8h)	STEL: 20 ppm 15 min STEL: 80 mg/m ³ 15 min TWA: 10 ppm 8 hr TWA: 40 mg/m ³ 8 hr Skin	TWA: 20 ppm (8 Stunden). AGW - exposure factor 2 TWA: 82 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 20 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time TWA: 82 mg/m ³ (8 Stunden). MAK can occur as vapor and aerosol at the same time Höhepunkt: 40 ppm Höhepunkt: 164 mg/m ³ Haut
Propylene carbonate			TWA: 2 ppm (8 Stunden). AGW - exposure factor 1 TWA: 8.5 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 2 ppm (8 Stunden). MAK TWA: 8.5 mg/m ³ (8 Stunden). MAK Höhepunkt: 2 ppm Höhepunkt: 8.5 mg/m ³
STOP SOLUTION - Sulphuric acid	TWA: 0.05 mg/m ³ (8h)	STEL: 0.15 mg/m ³ 15 min TWA: 0.05 mg/m ³ 8 hr	TWA: 0.1 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 0.1 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.1 mg/m ³

Exposure Controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

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Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance	Varies	
Physical State	Liquid	
	Lyophilized Pellets Solid	
Odor	No information available	
Odor Threshold	No data available	
pH	Not applicable	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	Not applicable	
Flash Point	Not applicable	Method - No information available
Evaporation Rate	Not applicable	Solid
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	Not applicable	Solid
Specific Gravity / Density	No data available	
Bulk Density	Not applicable	Liquid
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
WASH BUFFER -	-3.6	
Tris-hydrochloride		
SUBSTRATE -	-0.46	
N-Methyl-2-pyrrolidone (NMP)		
Propylene carbonate	-0.5	
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	Not applicable	Solid
Explosive Properties	No information available	
Oxidizing Properties	No information available	

SECTION 10: STABILITY AND REACTIVITY

Reactivity

None known, based on information available.

Chemical Stability

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Stable under recommended storage conditions.

Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.
None under normal processing.

Conditions to Avoid

None known.

Incompatible Materials

None known.

Hazardous Decomposition Products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects

Acute Toxicity

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	LD50 = 3914 mg/kg (Rat)	LD50 = 8 g/kg (Rabbit)	LC50 > 5.1 mg/L (Rat) 4 h
Propylene carbonate	LD50 = 29000 mg/kg (Rat)	LD50 > 3000 mg/kg (Rabbit)	
STOP SOLUTION - Sulphuric acid	LD50 = 2140 mg/kg (Rat)		LC50 = 0.375 mg/L (Rat) 4 h
Tris (hydroxymethyl) aminomethane	LD50 = 5900 mg/kg (Rat)	LD50 > 5000 mg/kg (Rat)	
1-Aminopyrine	LD50 = 285 mg/kg (Rat)		

Chronic Toxicity

Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	IARC	UK
STOP SOLUTION - Sulphuric acid	Group 1	

Legend:

X - Listed '-' - Not Listed R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

Sensitization

No information available

Mutagenic Effects

No information available

Reproductive Effects

No information available

Developmental Effects

May cause harm to the unborn child

Target Organs

No information available.

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SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. .

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
WASH BUFFER - Tris-hydrochloride		EC50 >100 mg/L/48h		
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	LC50: = 1400 mg/L, 96h static (Poecilia reticulata) LC50: = 1072 mg/L, 96h static (Pimephales promelas) LC50: = 832 mg/L, 96h static (Lepomis macrochirus)	EC50: = 4897 mg/L, 48h (Daphnia magna)	EC50: > 500 mg/L, 72h (Desmodesmus subspicatus)	
Propylene carbonate	Leuciscus idus: LC50: 5300 mg/L/96h	EC50: > 500 mg/L, 48h (Daphnia magna)	EC50: > 500 mg/L, 72h (Desmodesmus subspicatus)	EC50 > 10000 mg/L 17 h
STOP SOLUTION - Sulphuric acid	LC50: > 500 mg/L, 96h static (Brachydanio rerio)	EC50: 29 mg/L/24h		

Persistence and degradability

No information available

Bioaccumulative potential

No information available

Component	log Pow	Bioconcentration factor (BCF)
WASH BUFFER - Tris-hydrochloride	-3.6	No data available
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	-0.46	No data available
Propylene carbonate	-0.5	No data available

Mobility in soil

No information available.

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Dispose of in accordance with federal, state and local regulations Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

Contaminated Packaging

Dispose of in accordance with local regulations Dispose of this container to hazardous or special waste collection point.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

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SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

UN-No UN2796
Hazard Class 8
Packing Group II
Proper Shipping Name Sulphuric acid

Road and Rail Transport

UN-No UN2796
Hazard Class 8
Packing Group II
Proper Shipping Name Sulphuric acid

IATA

UN-No UN2796
Hazard Class 8
Packing Group II
Proper Shipping Name Sulphuric acid

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

Component	EINECS	TSCA	DSL	PICCS	ENCS	ISHL	IECSC	AICS	KECL
WASH BUFFER - Tris-hydrochloride	214-684-5	X	X	X	X		X	X	KE-34819
SUBSTRATE - N-Methyl-2-pyrrolidone (NMP)	-	X	X	X	X	X	X	X	KE-25324
Propylene carbonate	203-572-1	X	X	X	X	X	X	X	KE-23785
STOP SOLUTION - Sulphuric acid	-	X	X	X	X	X	X	X	KE-32570
Tris (hydroxymethyl) aminomethane	-	X	X	X	X	X	X	X	KE-01403
1-Aminopyrine	-	X	-	X	X	X	-	X	-

Component	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
STOP SOLUTION - Sulphuric acid				Annex I - Y34

National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance
Ozone Depletion Potential This product does not contain any known or suspected substance

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SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

28-Mar-2023

Revision Summary

SDS sections updated, 3.

In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

Disclaimer

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End of Safety Data Sheet